A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 15

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1 Diversity:** HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change **1** No single source could meet demand all year – battery storage on the island grid smooths fluctuations **1** Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife **1** Wood biomass heating – insulation such as straw-bale housing – some electric vehicles **1 Habitat protection and creation:** fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow) – eagles and hen harriers nesting **1** Recyclables shipped off – garden / food waste composted – low-impact food production – beach litter collection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Small groups can organise, agree and act quickly – decisions benefit people who see the results directly **1** Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Renewable energy is abundant but geographically and temporally variable – a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand **1** DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Local people had used the area for fuelwood, timber for house building, clearings for farming, hunting and the collection of medicines and fibres, all of which protection could have stopped **1** Local people are employed as rangers, forestry workers and in ecotourism, so the forest is worth more to them standing than cleared **1** Ecotourist groups eat in local restaurants, are fed on local produce and buy locally made souvenirs, spreading the income through the community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Trees are only felled where other trees of the same species stand upwind of the clearing, so their seed will blow in and recolonise it **1** Small trees are left standing so that they can grow and be harvested in the future when they are larger **1** Selected large trees are left because they produce many seeds that fill the gaps left by the trees that were felled **1** No mechanical or harvesting activity takes place near rivers, because aquatic ecosystems are ecologically sensitive **1** The use of large machines is restricted because they damage the trunks and roots of the trees that are not being felled **1** Track routes are planned to avoid important trees and their roots, including rare trees and those that feed monkeys or birds, and old tracks are re-used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** The tag allows the timber to be tracked from the stump where it grew to its final retail destination, such as a DIY store in the UK **1** That chain of custody is the evidence the FSC needs that the wood really came from a sustainably managed forest and was not mixed with illegally logged timber **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Ecotourism:** visitors pay to see the forest and its wildlife, and local businesses supply them **1** Timber confiscated from illegal loggers by the rangers is sold, and the money pays for conservation work **1** Electricity companies, airlines and individuals fund an area of forest as a carbon sequestration research project, because they want to offset atmospheric carbon dioxide **1** Lower-value species such as zericote are developed for furniture, musical instruments and carving, and small branches and offcuts are given to local artisans **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The canopy is only opened in small gaps, so the forest microclimate - shade, humidity and temperature - is maintained and shade-dependent species survive **1** The soil keeps its cover of vegetation and leaf litter, and the roots that bind it, so erosion, leaching and sedimentation of rivers are far lower **1** Seed trees and small trees remain, so the gaps are recolonised by the same species and the forest regenerates naturally without replanting **1** Habitats and food sources are left in place and the forest is not fragmented, so populations of birds and mammals are not isolated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369, 386-387; AQA spec 3.5.3.3.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** The new opportunity is overwhelmingly attractive (economic benefit) **1** Lack of knowledge / understanding of the impacts **1** The impacts occur elsewhere or much later, so are not evident to those who benefit **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cultural belief that resources should be exploited to satisfy needs only **1** Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Resources are brought in from other areas **1** If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**1 mark each** 1 mark for each of the 3 points below. **1** Easier access to energy has made many new activities possible **1** The scale of activities has greatly increased **1** Toxic wastes are produced that cannot be decontaminated by natural processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	
04.1	1 mark for the point below. **1** Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Zebu tolerate heat but have a low milk yield – Ayrshire have a high milk yield but are not heat tolerant **1** Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Replacement of valuable animals that die with genetically identical individuals **1** Replacement of herds culled during a disease outbreak **1** Production of large numbers of individuals with desirable characteristics **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
05.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Hydrological cycle **1** Carbon cycle **1** Nitrogen cycle **1** Atmospheric circulation **1** Thermohaline circulation **1** Photosynthesis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	2
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Enzymes reduce the activation energy of reactions **1** So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2
05.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Haber process (nitrogen fixation) **1** Incineration of wastes (decomposition) **1** Many manufacturing processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **Incineration** → burning waste, usually with energy recovery	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Overall decline from 62% to 22% **1** Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** The polyps expel their symbiotic zooxanthellae so the coral loses colour – bleached corals are alive but grow very slowly and may die if algae do not return **1** Raised water temperature (climate change / El Niño) **1 Pollution:** sewage, pesticides, cyanide, sunscreen – low oxygen from zooplankton blooms after overfishing **1** High light / solar activity – ocean acidification hinders skeleton formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiotic** → a close relationship in which both species benefit **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps **1** Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release **1** Physical damage from anchors, divers, souvenir collection – loss of mangroves / seagrass nurseries that trap sediment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
06.4	1 mark for the point below. **1** Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1
07.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** Mars has low mass so weak gravity and could not retain a dense atmosphere **1** Very low atmospheric pressure means water would boil / evaporate away even at low temperatures **1** Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO2, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil **1** Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured **1** Too close (Venus) - too hot – too far (Mars) - too cold **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface **1** The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun **1** Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted – the Moon's gravity keeps the tilt stable and creates tides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds **1** Water-repelling surface of lotus flowers **1** Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to **1** Hooked burrs on seeds that stick to animal fur **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** The mound absorbs sunlight and warms – hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) **1** Air blowing over the mound has a lower pressure which also draws air out **1** Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Marsupials:** young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems **1** Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns **1** Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
09.4	1 mark for the point below. **1 They are light but strong:** hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (15 - 6) / 6 x 100 **1** = 150% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cells shed by the newts into the water contain their DNA **1** Water samples are analysed for DNA sequences unique to the species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** Catches depend on trapping effort, weather and newt activity on the night, so they vary from year to year **1** Trapping only samples one pond and one part of the population, whereas eDNA surveys all 20 ponds with a standard method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Image databases of unique markings (tiger stripes, whale shark spots, dolphin fins) giving territory size, movements, lifespan and social groups – limited to species with unique features • Motion-sensitive and infrared cameras, CCTV of nests with less disturbance – marking (rings, wing tags) • **DNA:** eDNA to detect presence (great crested newts, invasive carp) – DNA databases to identify regional populations and trace timber, ivory and fish • **Auditory monitoring:** bat detectors and sonograms for presence, abundance and activity • **Position monitoring:** radio, GPS and acoustic transmitters – geolocator tags for small birds – data recorders – carrier systems - ROVs / drones, AUVs, balloons, aircraft, satellites, animals • **Satellites:** visible light for land use and deforestation, infrared for vegetation and temperature, radar for wind, waves, sea level and oil – GPS for tracking • **Evaluation:** greater coverage, less disturbance and continuous data vs cost, battery / weight limits, calibration, need for ground-truthing, ethical issues of tagging – indirect evidence and traditional methods still needed • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	9
	Do not credit • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	New technologies have transformed what can be monitored and how often. Image databases of unique markings - tiger stripes, whale-shark spots, dolphin fin scars - let individuals be recognised, giving territory size, movements, lifespan and social structure without repeated capture, though only for species with unique features. Motion-sensitive and infrared cameras and CCTV record activity day and night with far less disturbance than observers, especially at nests.	
	Genetics adds detection where nothing can be seen: eDNA in water reveals great crested newts or invasive carp from shed cells, and DNA databases identify which regional population timber, ivory or fish came from, supporting enforcement. Bat detectors and sonograms give presence, abundance and activity of species heard but not seen. Position monitoring uses radio, GPS and acoustic transmitters, or tiny geolocator tags for birds too small to carry a transmitter, while data recorders on drones, AUVs, balloons, aircraft, satellites or the animals themselves collect abiotic data. Satellites map land use and deforestation in visible light, vegetation density and temperature in infrared, and wind, waves, sea level and oil slicks with radar that works through cloud.	
	The gains are coverage, continuity and reduced disturbance. The limits are cost, battery life and weight, the need to calibrate sensors and ground-truth remote data, ethical questions about tagging, and the fact that many species still require traditional methods and indirect evidence. New technology therefore complements rather than replaces fieldwork, but for rare, remote or nocturnal species it has made effective conservation possible where it was not before.	
	Weaker answer - Level 1 (3 marks)	
	New technologies help monitor wildlife. Cameras can take pictures of animals at night. GPS collars can track where animals go. Satellites can see deforestation from space. DNA can tell which animals are present. Bat detectors can hear bats. These help conservation because we know where animals are and how many there are. They can be expensive though.	
	What makes the difference	
	• Level 1 lists technologies with no detail of how they work or what data they give; Level 3 links each to the information it provides (territory size, presence, activity).	
	• Level 3's evaluation considers several limitations (calibration, ground-truthing, ethics, species coverage) and reaches a balanced judgement; Level 1 has one word - 'expensive'.	
	• Structure: Level 3 groups by imagery, genetics, sound, position and remote sensing; the list in Level 1 has no organisation.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations – positive feedback and tipping points caused by humans • **Energy:** solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process) – renewable energy, heat pumps, bioleaching, fuel cells • **Materials:** cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals – recycling, Cradle to Cradle, industrial symbiosis • **Diversity and stability:** rainforests / reefs vs agroecosystems – diverse energy mixes • Optimum vs maximum production – supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics – limits and judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Marine productivity:** light, nutrients, upwelling – where fisheries occur • **Methods:** demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • **Impacts:** overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • **Population dynamics and MSY:** Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • **Management:** quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear – EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover) – role of aquaculture	25
	– judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	

Do not write outside the box